



Using a Different Tracer to Measure Outflow Opening Angles of Forming Stars

Wyatt Ulsh, Michael Dunham (Department of Physics and Astronomy)



wulsh@middlebury.edu

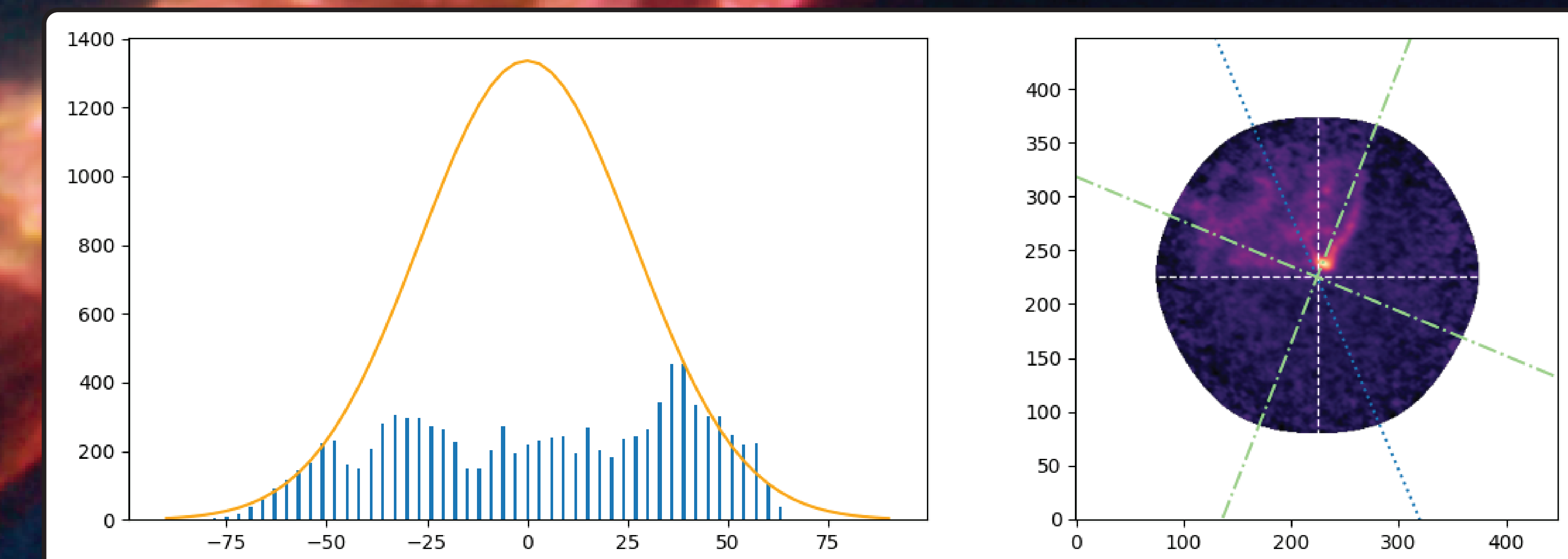
Background

- Approximately two-thirds of mass is ejected during stellar formation.
- One proposed mechanism for this mass ejection is bipolar molecular outflows.
- To test this, we see if the outflow opening angles widen enough with time to remove this much mass.
- Typically, ^{12}CO is used to trace these outflows because it is relatively abundant and therefore easy to observe.
- Instead, we are using ^{13}CO as tracer because it may provide a more accurate measurement.

Project Details

- ^{13}CO is approximately two orders of magnitude less abundant than ^{12}CO , so it appears much dimmer in images and requires longer observation times.
- The radio interferometer images we are working with were not taken specifically for ^{13}CO , so the images are very dim and difficult to extract details from.
- Due to projection issues, some of the sources do not appear to be centered where the outflows would suggest they should be. This means that the opening angles that we measure may not have physical significance for some images. Since we are comparing the same image across different wavelengths, the difference between the opening angles still holds significance.

Measurement Method



HOPS-11 Redshifted ^{13}CO Gaussian Fit and Image

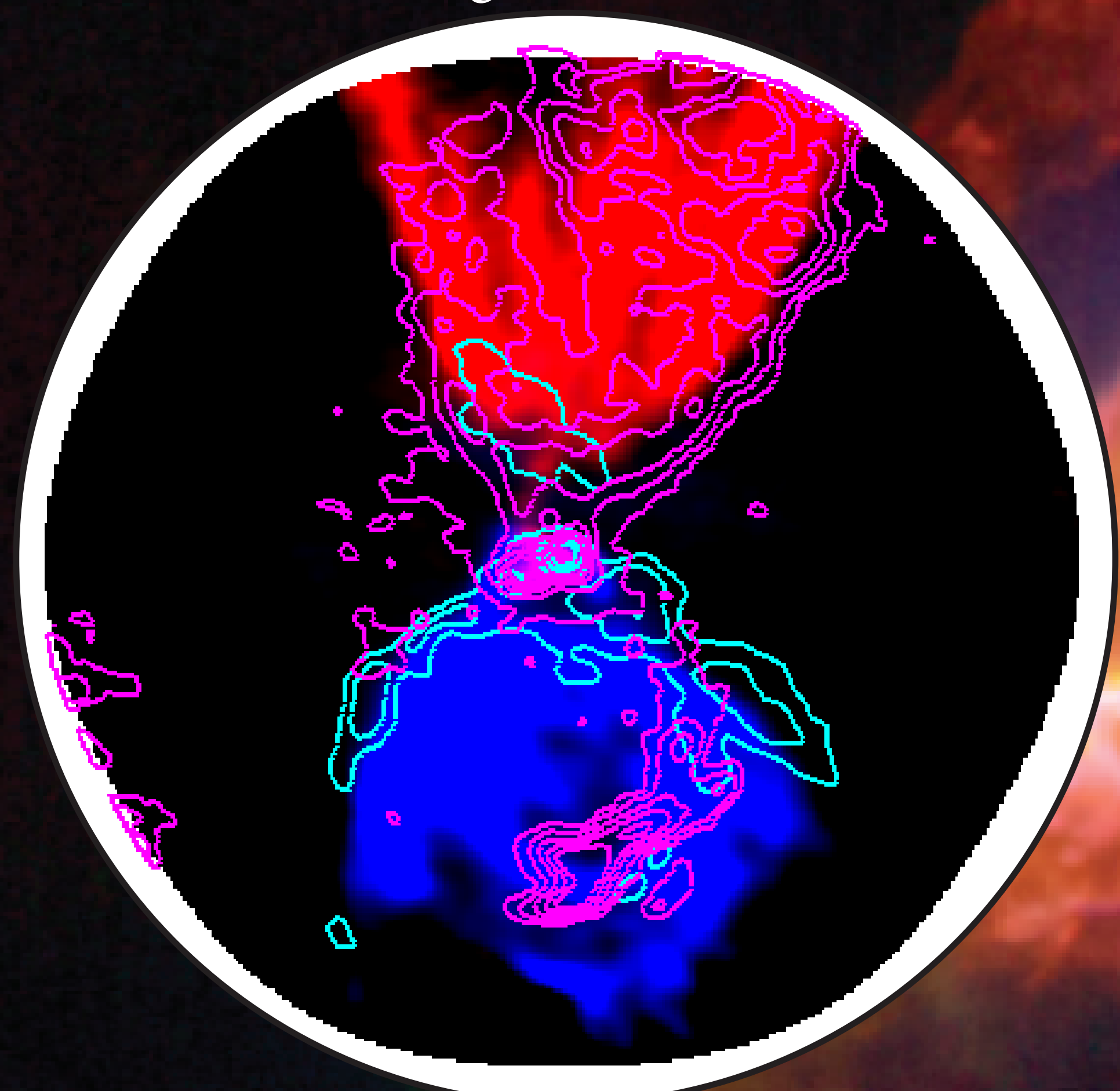
The Gaussian graph has units of number of pixels on y-axis and degrees on the x-axis. Bins of 3 degrees are used to create the distribution that we fit the Gaussian to.

Preliminary Results

Though much more analysis is needed, we have so far found that most ^{13}CO images are approximately 10 degrees wider than their corresponding ^{12}CO images. More measurements are underway.

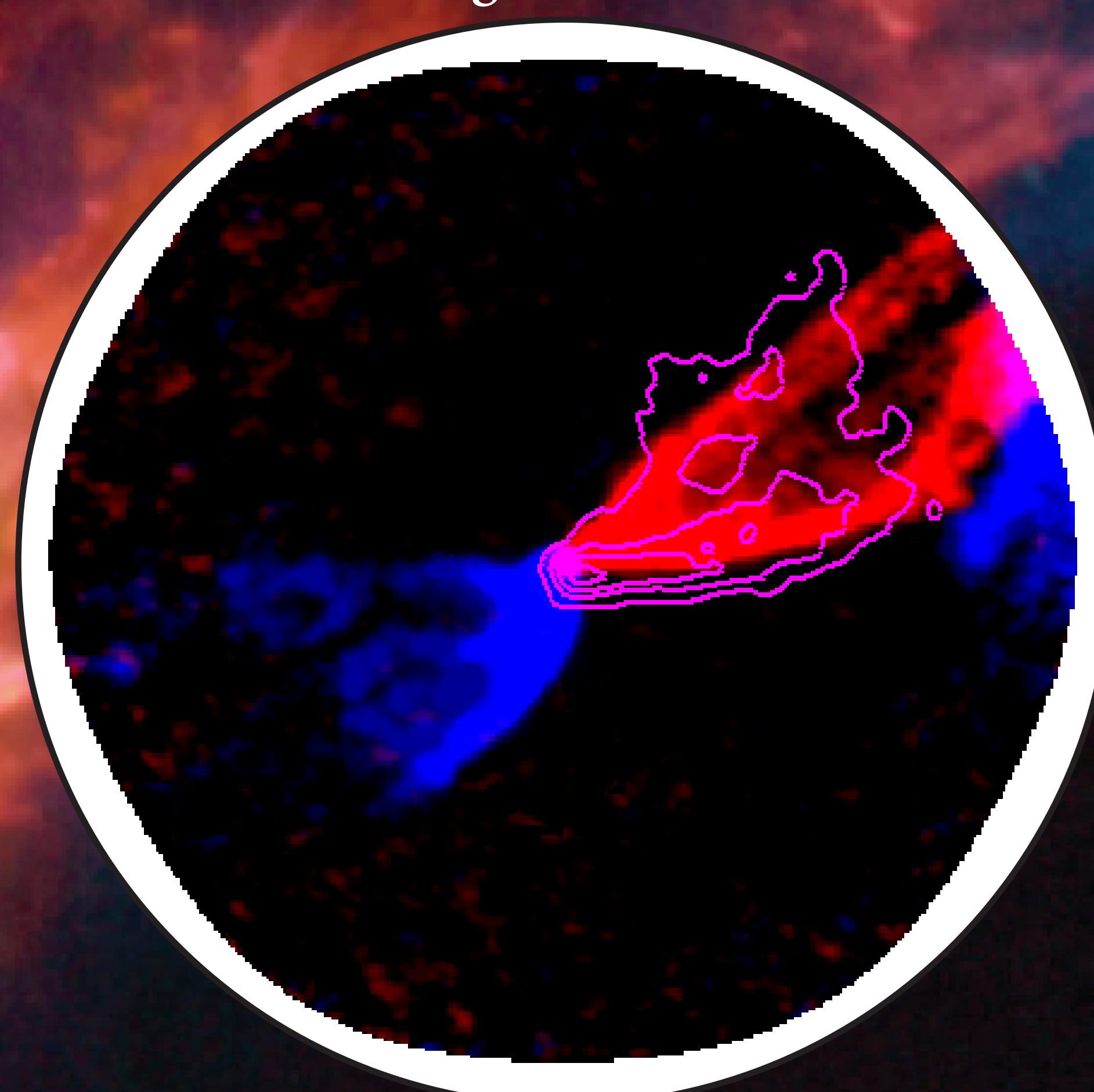
HOPS-177

^{13}CO contours plotted over ^{12}CO integrated intensities



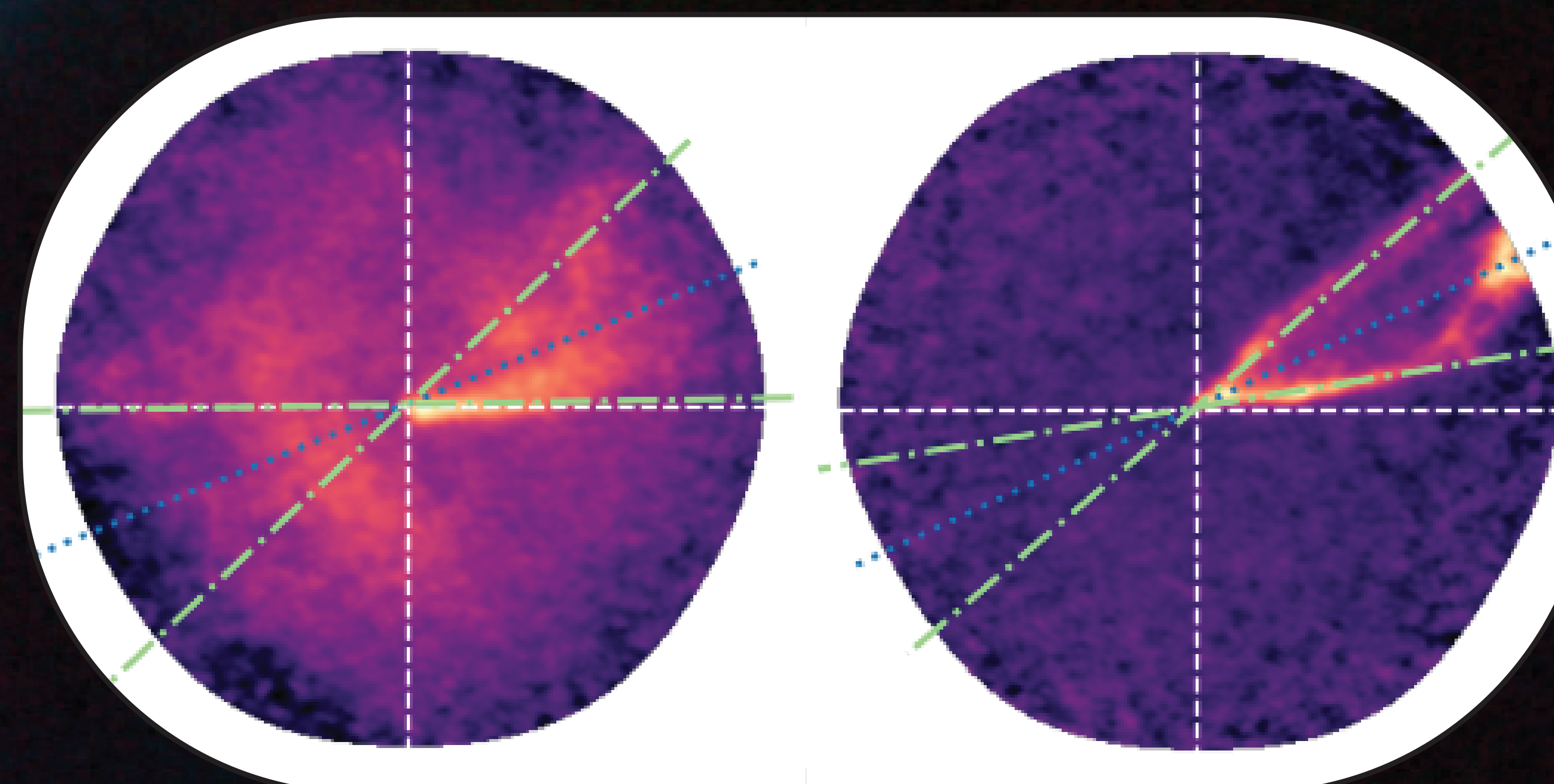
HOPS-135

^{13}CO red contours plotted over ^{12}CO integrated intensities



Redshifted images of HOPS-135

^{13}CO on left, ^{12}CO on right (10 degree difference)
These are the images the analysis code creates.



This is the method we used to measure the outflow opening angles that eliminates as much subjectivity as possible:

1. Create an integrated intensity (moment 0) map over the redshifted and blueshifted velocity ranges where outflow emission is detected.
2. Create a binary pixel map, giving a value of 1 to pixels above a threshold value determined by the background noise of the integrated intensity map.
3. Enter a guess of the position angle of the central axis of the outflow.
4. For each pixel of value 1 in the pixel map, measure the angle between the pixel and the central axis.
5. Fit a Gaussian to the distribution of pixel angles.
6. Adjust the position angle so that the Gaussian is centered at 0 degrees.
7. The full width at quarter maximum of the Gaussian is the final measurement of the opening angle (Offner et al. 2011)

The image above is an output of the code we wrote that performs these measurements.